

VR-Sets. Preprint.

Part V. VR Numbers in VR-Sets

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V.0. Introduction

This part of the preprint establishes the correspondence between the VR numbers — the von Neumann ordinals O_n , obtained in the VR system (Reznik, 2026) — and the operational sets of VR-Sets. The aim is modest: to show that the VR numbers are specific operational sets; that they fall automatically into the ZFC-mode as a theorem (not as a restriction); and that the arithmetic developed in VR (Part I §7) is inherited in VR-Sets via direct isomorphism, without the need for re-proof.

The part is short by design. The substantive work on the axiomatisation of arithmetic has already been done in VR; the present part only specifies how this arithmetic looks in the new ontology. No new arithmetic theorems are proved here.

Status reminder. Throughout this part the phrase “belongs to the ZFC-mode” is used in the strict technical sense established in Part IV: a set is well-founded and operationally describable. This is Theorem IV.1 territory and does not depend on Conjecture IV.1 (mutual interpretability of the ZFC-mode with a countable model of classical ZFC), which remains open. Accordingly, when this part says that VR numbers and the apparatus of VR-Numbers are inherited in VR-Sets without re-proof, the inheritance is from VR and from VR-Numbers themselves — via the operational isomorphism established below — not from classical ZFC.

V.1. VR numbers as operational sets

In VR (Part I, Definitions 4–5) the von Neumann ordinals are defined recursively:

$$O_0 := \emptyset, \quad O_{n+1} := t(O_n) = O_n \cup \{O_n\}.$$

Substantively:

$$O_0 = \emptyset,$$

$$O_1 = \{\emptyset\},$$

$$O_2 = \{\emptyset, \{\emptyset\}\},$$

$$O_3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\},$$

...

In VR-Sets each of these expressions is interpreted as the description of an operational functionality. Formally:

Definition V.1 (VR numbers in VR-Sets)

The operational set O_n is the functionality whose description is as follows: "reveal in turn $O_0, O_1, \dots, O_{n-1}$; on subsequent queries reveal nothing." Recursive base: $O_0 \equiv \emptyset$ (the empty operability, revealing nothing).

Commentary. This is precisely what the formula $O_{n+1} = O_n \cup \{O_n\}$ says: the elements of O_{n+1} are all elements of O_n (that is, $O_0, \dots, O_{n-1}$) plus O_n itself. In the operational formulation this means that the functionality O_{n+1} reveals $O_0, \dots, O_{n-1}, O_n$ — $n+1$ elements in all.

The operational depth of O_n equals n (Lemma 3 of Part II): the unfolding $O_n \rightarrow O_{n-1} \rightarrow O_{n-2} \rightarrow \dots \rightarrow O_0 = \emptyset$ terminates in exactly n steps.

V.2. ω as an operational set

The set of natural numbers ω in VR-Sets is the functionality already introduced in Part II with the description:

Definition V.2 (ω)

ω is the operational set whose functionality, on the n -th query, reveals O_n . The functionality does not terminate at any finite step (it is infinite by Definition 3 of Part II) and is non-cyclic (ω is not among the revealed elements).

The existence of ω was proved in Part III §III.6 as a closure theorem: ω is operationally describable (on every query the response is defined — a specific O_n), and by the closure principle such a functionality is a set.

This accords with the axiom of infinity in its operational reformulation: ω in VR-Sets is not a "completed totality of all naturals" but an unboundedly applicable operation of enumerating O_n . The substantive divergence with classical ZF at this point is methodological, not mathematical: mathematically ω in VR-Sets is isomorphic to ω in classical ZF (see §V.4).

V.3. VR numbers in the ZFC-mode: theorem

The principal technical result of this part is that VR numbers fall automatically into the ZFC-mode of VR-Sets. This is not a restriction imposed on the construction, but a theorem about a structural property of the numbers.

Theorem V.1

Every VR number — O_n for arbitrary $n \geq 0$, as well as ω itself — is a well-founded operational set. Accordingly, all VR numbers belong to the ZFC-mode of VR-Sets.

Proof

Part 1: every O_n is well-founded. Induction on n .

Base ($n = 0$). $O_0 = \emptyset$ is well-founded trivially: $\emptyset$ has no elements, no unfoldings, and the termination requirement holds vacuously.

Step ($n \rightarrow n+1$). Suppose $O_0, O_1, \dots, O_n$ are all well-founded. We show that O_{n+1} is well-founded. Any successive unfolding starting with O_{n+1} yields on the first step one of the elements of O_{n+1} , that is, one of $O_0, \dots, O_n$. By the inductive hypothesis each of these is well-founded — and hence the chain of unfolding from that element terminates at $\emptyset$ in a finite number of steps. Consequently every unfolding of O_{n+1} terminates. O_{n+1} is well-founded.

By induction (axiom A4 from VR, lifted to the level of sets through Lemma 3 of Part II) all O_n are well-founded.

Part 2: ω is well-founded. Any successive unfolding starting with ω yields on the first step some element of ω — namely a specific O_n with finite n . By Part 1, O_n is well-founded; the unfolding chain from O_n terminates at $\emptyset$ in n steps. Consequently the chain of unfolding of ω terminates in $n+1$ steps. ω is well-founded.

Conclusion. By Definition IV.3 (Part IV), a set belongs to the ZFC-mode if it is operationally describable and well-founded. All O_n and ω are describable (Part III §III.6 for ω ; for O_n — by the construction of Definition V.1) and well-founded (Parts 1 and 2 of this proof). All VR numbers belong to the ZFC-mode.

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Remark: correspondence with VR Part II.5

In VR (Part II §5) it was proved: $t(x) \neq x$ for every object of the system, and the chains of membership in the objects of VR are finite. These facts are reformulations of well-foundedness in the language of VR. Theorem V.1 is the same content, expressed in the language of VR-Sets:

- $t(x) \neq x$ in VR $\leftrightarrow$ O_{n+1} is not cyclic on itself in VR-Sets (the unfolding of O_{n+1} yields elements of strictly smaller depth, not O_{n+1} itself).
- Chains of membership in VR are finite $\leftrightarrow$ successive unfoldings in VR-Sets terminate at $\emptyset$.

No new proof was carried out; Theorem V.1 is a transcription of what has already been proved into the new language. This is consistent with the general methodological position: VR-Sets does not re-prove arithmetic, it interprets it.

V.4. Isomorphism with the natural numbers of VR

The VR numbers in VR-Sets are precisely the VR numbers of the original VR system, presented operationally. We fix this formally.

Theorem V.2 (isomorphism)

The mapping $\varphi: \mathbb{N}_{VR} \rightarrow \mathbb{N}_{VR\text{-Sets}}$, sending each VR number O_n of the original system to the corresponding operational functionality O_n in VR-Sets, is an isomorphism of structures:

- (a) *bijectivity*: to each VR number there corresponds exactly one operational functionality in VR-Sets, and conversely;
- (b) *preservation of succession*: $\varphi(t(O_n)) \equiv t(\varphi(O_n))$ in VR-Sets;

- (c) *preservation of membership*: $O_m \in O_n$ in VR $\iff \varphi(O_m) \in \varphi(O_n)$ in VR-Sets;
- (d) *preservation of arithmetic operations*: addition, multiplication, and exponentiation defined in VR Part I §7 agree with the corresponding operations on operational functionalities in VR-Sets.

Proof

- (a) Bijectivity by construction: φ is defined recursively through O_n in both systems; the recursion determines the correspondence uniquely.
- (b) Preservation of succession. In VR succession is given by axiom A3: $t(x) = x \cup \{x\}$. In VR-Sets, succession from O_n is the operational functionality revealing $O_0, \dots, O_{n-1}, O_n$ (that is, everything that O_n reveals, plus O_n itself). This corresponds precisely to the union of the set of elements of O_n and $\{O_n\}$ — that is, $t(O_n) = O_n \cup \{O_n\}$ in VR.
- (c) Preservation of membership. In VR, $x \in y$ means that x is one of the objects mentioned in the construction of y by A3. In VR-Sets, $x \in y$ means that the functionality y reveals x as one of its elements. By the construction of Definition V.1 these two relations coincide.
- (d) Preservation of arithmetic operations. Addition, multiplication, and exponentiation in VR are defined recursively through t (Definitions 7–9 of Part I of VR). These definitions carry over into VR-Sets literally: they operate through succession and the base O_0 , and both constructions are coordinated between the systems by clauses (a), (b). All arithmetic theorems T1–T4 of VR Part I §7 (commutativity, associativity, distributivity, $O_1 + O_1 = O_2$) are inherited in VR-Sets without re-proof.

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Corollary V.1

The entire arithmetic developed in VR Part I — a Peano-equivalent theory of natural numbers with all standard operations and theorems — is available in VR-Sets through the isomorphism φ without the need for additional work.

Methodological commentary. This result is the consequence of a deliberate methodological decision: VR-Sets does not replace VR but interprets it. The axiomatic work justifying arithmetic was carried out in VR once; in VR-Sets, arithmetic appears as an already-prepared structure, presented operationally. This is consistent with the general logic of the cycle of works: VR provides the minimal axiomatic; VR-Numbers extends it to $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$; VR-Sets places all of this in an ontological context. No part of the cycle duplicates the work of another.

V.5. Extension to VR-Numbers

We briefly fix that not only the natural numbers, but the whole superstructure of VR-Numbers ($\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$) is inherited in VR-Sets through the same mechanism.

General principle

In VR-Numbers (Reznik, 2026) the numerical extensions are built as follows:

- $\mathbb{N}$ — the von Neumann ordinals O_n (ontologically — the sole level with sets proper as objects);
- $\mathbb{Z}, \mathbb{Q}$ — operational syntax over $\mathbb{N}$ (not pair-objects, but operational procedures with order built into the description);
- $\mathbb{R}$ — algorithms computing real approximations (a countable subfield of the classical $\mathbb{R}$, isomorphic to the field of computable reals);
- $\mathbb{C}$ — a two-axis structure motivated by the duality of axiom A1 (the two implications $F \rightarrow T$ and $T \rightarrow F$), with algebraic gluing through $i^2 = -1$.

All these extensions are **ontologically minimal**: none of them introduces new ontological primitives beyond $\emptyset$. Integers, rationals, reals, complex numbers are operational procedures of varying complexity over natural numbers, which are themselves operational sets.

Transition into VR-Sets

Since all extensions of VR-Numbers ontologically reduce to operational procedures over natural numbers, and since natural numbers are represented in VR-Sets through the isomorphism ϕ (Theorem V.2), the whole apparatus of VR-Numbers transfers into VR-Sets automatically:

- $\mathbb{Z}, \mathbb{Q}$ in VR-Sets are operational sets whose descriptions use elements of $\mathbb{N}_{\text{VR-Sets}}$ as basic building material. All operational procedures of VR-Numbers (addition of integers via signed descriptions, rationals as numerator/denominator pairs given operationally, and so on) are applicable directly.
- $\mathbb{R}_{\text{VR-Sets}}$ is a countable subset of the ZFC-mode containing operational sets representing computable algorithms. Each such set is well-founded (the elements of algorithms — rational approximations — are well-founded; the description of an algorithm is finite, not cycled on itself).
- $\mathbb{C}_{\text{VR-Sets}}$ is the two-axis structure over $\mathbb{R}_{\text{VR-Sets}}$, preserving the algebraic relation $i^2 = -1$.

No new theorems are proved here. The aim of this section is to fix that the transition VR-Numbers $\rightarrow$ VR-Sets is as seamless as the transition VR $\rightarrow$ VR-Sets. A detailed operational description of each extension is given in VR-Numbers; here we only assert transferability.

V.6. Under the standard coding, VR numbers do not use the ZFA-mode

One structural fact deserves explicit mention: under the standard von Neumann coding of numbers used in this preprint, the VR numbers fall entirely within the ZFC-mode. The ZFA-mode — self-references, cycles, non-well-founded sets — is neither needed nor used for arithmetic in this coding. Alternative non-von-Neumann codings (including coinductive constructions in the ZFA-mode) are possible in principle and are noted as an open direction in Part IX, Extension 4; this assertion concerns only the standard coding.

Assertion V.1

No VR number — neither natural (O_n or ω), nor integer, nor rational, nor real, nor complex — is, under the standard coding used in this preprint, a cyclic or non-well-founded set.

Justification. For natural numbers and ω this is Theorem V.1. For $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$: the operational procedures defining these numbers in VR-Numbers operate through finite syntactic records over already-constructed natural numbers; none of these records refers to itself; consequently the describable functionalities are well-founded.

Methodological significance. This is consistent with the general programme of VR: classical mathematics — arithmetic, analysis, algebra — lives entirely in the ZFC-mode of VR-Sets. The ZFA-mode (the theory of non-correspondence, circular structures, situational semantics) is a separate area of application that does not intersect with numbers. The distinction between modes in VR-Sets is therefore not "different mathematics" but "different applications of a single ontology": classical mathematics of numbers uses the ZFC-mode as a theorem about a structural property of its objects; mathematics of self-referential structures uses the ZFA-mode as an explicit ontological choice.

V.7. Summary and transition to Part VI

This part has established the following:

- (1) **VR numbers are specific operational sets of VR-Sets.** Each O_n is a functionality with operational depth n ; ω is a functionality with non-finite but non-cyclic depth (Definitions V.1 and V.2).
- (2) **VR numbers automatically belong to the ZFC-mode** (Theorem V.1). This is not a restriction but a theorem about a structural property — the analogue of what in VR Part II.5 was expressed as $t(x) \neq x$ and the finiteness of chains of membership.
- (3) **Isomorphism with VR** (Theorem V.2): the VR numbers in VR-Sets are precisely the VR numbers of the original system, presented operationally. The whole arithmetic developed in VR is inherited without re-proof (Corollary V.1).
- (4) **VR-Numbers transfers analogously:** $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are operational procedures over natural numbers, all well-founded, all in the ZFC-mode (§V.5).
- (5) **Numbers do not use the ZFA-mode:** the whole classical mathematics of numbers lives in the ZFC-mode (§V.6).

What follows. Part VI unfolds the central divergence of VR-Sets with classical ZFC — the countability of the operational universe. Already in the present part it is visible that VR numbers live on a countable structure: $\mathbb{N}_{\text{VR-Sets}}$ is countable, $\mathbb{R}_{\text{VR-Sets}}$ is countable (isomorphic to the field of computable reals). Part VI establishes the countability of the entire operational universe in full detail, analyses Cantor's diagonal argument in the operational ontology, proves AC as a theorem on the countable universe, and fixes the absence of the Banach–Tarski paradox by construction.